

Notes: Baker, Gibbons, and Murphy (QJE, 1994)

Subjective Performance Measures in Optimal Incentive Contracts

Contents

1	Big picture	2
2	Incremental contribution of the paper	2
3	Data, institutional motivation, and model environment	3
3.1	No empirical dataset: theory motivated by organizational practice	3
3.2	Economic actors and timeline	3
3.3	Objective and subjective performance measures	3
3.4	Worker preferences and compensation	4
4	Benchmark models and main theoretical specifications	4
4.1	Benchmark 1: explicit contract based on the objective measure	4
4.2	Benchmark 2: implicit contract based on subjective performance	4
4.3	Combining explicit and implicit contracts: Case 1, profitable explicit fallback	5
4.4	Case 1 comparative statics	5
4.5	Combining explicit and implicit contracts: Case 2, unprofitable explicit fallback	5
4.6	Imperfect subjective performance assessment	6
4.7	Subjective weights on objective measures	6
5	Theoretical identification strategy	6
5.1	Step 1: identify the objective-measure distortion	6
5.2	Step 2: identify the self-enforcement constraint	6
5.3	Step 3: compare contracts in isolation and in combination	7
5.4	Step 4: map parameter changes to organizational predictions	7
6	Figures and detailed interpretations	8
6.1	Figures I and II: subjective bonus feasibility with discount rate and alternative wage	8
6.2	Figures III and IV: substitutability and complementarity between explicit and implicit incentives	8
6.3	Figures V and VI: subjective weights on objective performance measures	9
7	Interpretive synthesis	10
7.1	Why objective measures can be too good	10
7.2	Why hybrid incentive systems are common	10
7.3	Why subjective pay is not simply favoritism	10
7.4	Connection to firm boundaries	10
8	Short summary	10

9	Conclusion	11
9.1	Core conclusion	11
9.2	Economic intuition	11
9.3	Incremental contribution in one sentence	11
9.4	Final takeaway	11

1 Big picture

- **Question.** How should firms combine objectively measured pay-for-performance with subjective performance assessment when objective performance measures are imperfect and subjective assessments cannot be court-enforced?
- **Main puzzle.** Firms often rely heavily on subjective bonuses even when they also observe objective measures such as output, revenues, fees, or trading profits. The paper asks when this mixture is optimal.
- **Core tension.** Objective measures are enforceable but distortionary: workers can game them. Subjective measures can better capture true contribution to firm value, but they require trust because firms can renege ex post.
- **Model form.** The paper combines two benchmark models: Baker’s explicit-contract model with a distorted objective signal and Bull-style repeated-game implicit contracts enforced by the firm’s reputation.
- **First main result.** Objective and subjective contracts can be substitutes in a strong feasibility sense: if objective contracts are sufficiently effective, the firm has an attractive fallback after reneging, so no implicit contract can be self-enforcing.
- **Second main result.** Objective and subjective contracts can also be complements: in some parameter regions neither an explicit contract nor an implicit contract alone is profitable, but their combination is profitable and incentive-compatible.
- **Third result.** Subjective weights on objective measures can mitigate the dysfunctional behavior created by purely explicit piece rates. The firm may deliberately leave the contract vague about how an observed objective result will be rewarded.
- **Economic takeaway.** A good incentive system is not simply “more pay-for-performance.” It must balance objective enforceability, incentive distortion, subjective information, and the value of a long-run relationship.

2 Incremental contribution of the paper

- **Formalizes subjective performance evaluation.** Earlier agency theory mainly emphasized objective but noisy measures. This paper explicitly models subjective evaluations as useful but noncontractible information.
- **Integrates explicit and implicit contracts.** It brings together the explicit-contract distortion logic of Baker (1992) and the self-enforcing relational-contract logic of Bull (1987) in a single framework.
- **Shows that explicit contracts can destroy implicit-contract feasibility.** The paper’s most distinctive insight is that an improved spot-market or objective-contract fallback can reduce the value of reputation and thereby make subjective bonuses impossible.
- **Shows that explicit and implicit contracts can be complements.** The paper also explains why combinations of objective piece rates and subjective bonus systems can succeed even when either system alone fails.
- **Explains deliberate contractual vagueness.** It provides a theory for why firms may specify how performance is measured but leave the reward for that performance partly subjective.
- **Connects formal contract theory to actual compensation practice.** The paper rationalizes examples such as investment banking bonuses and Lincoln Electric’s combination of piece rates and subjective bonuses.
- **Extends the logic beyond compensation.** The same trade-off applies to promotion, job security, at-will employment, and other employment practices that combine enforceable and relational elements.

3 Data, institutional motivation, and model environment

3.1 No empirical dataset: theory motivated by organizational practice

This is a theory paper. It does not assemble a new dataset or estimate reduced-form treatment effects. Instead, it uses case examples and institutional observations to motivate a formal model of incentive contracting.

- **Dysfunctional objective incentives.** Heinz, Dun & Bradstreet, and Sears illustrate that employees can respond to incentive formulas in ways that raise measured performance while reducing long-run firm value.
- **Subjective assessment in practice.** Investment banks use subjective bonuses to reward deal quality, client satisfaction, training, marketing, and other dimensions that are not captured by simple objective measures.
- **Hybrid incentive systems.** Lincoln Electric combines objective piece rates with subjective bonuses based on cooperation, innovation, dependability, and other qualitative dimensions.

Interpretation: The paper’s “data” are organizational facts that standard principal-agent models struggle to explain: objective measures are often available, yet firms still rely on subjective evaluations and discretionary bonuses.

3.2 Economic actors and timeline

The model is a repeated game between one firm and one worker. In each period:

1. the firm offers a compensation contract;
2. the worker privately observes information about the distortion in the objective measure;
3. the worker chooses unobservable action $a \in [0, 1]$;
4. the worker’s contribution to firm value $y \in \{0, 1\}$ and objective performance measure $p \in \{0, 1\}$ are realized;
5. explicit compensation is enforced by courts, while subjective bonuses must be honored voluntarily by the firm.

Interpretation: The chronology separates two frictions. First, the worker privately understands how the objective measure can be manipulated. Second, the firm cannot legally commit to subjective payments, so subjective pay requires self-enforcement.

3.3 Objective and subjective performance measures

The worker’s action is the probability of contributing one unit to firm value:

$$\Pr(y = 1 \mid a) = a.$$

The objective measure p is a distorted proxy. The worker privately observes $\mu > 0$, and

$$\Pr(p = 1 \mid a, \mu) = \mu a, \quad E[\mu] = 1.$$

The variance $\text{var}(\mu)$ measures the distortion of the objective measure.

- If $\mu > 1$, the objective measure is too easy to improve relative to firm value.
- If $\mu < 1$, the objective measure is too hard to improve relative to firm value.
- If $\text{var}(\mu)$ is high, an objective piece rate creates strong state-contingent distortions.

Interpretation: The objective measure is unbiased on average but distorted in states. This is why objective incentives can be attractive and dangerous at the same time.

3.4 Worker preferences and compensation

The worker receives salary s , an implicit bonus b tied to the subjective assessment, and an explicit piece rate β tied to the objective measure. The cost of action is quadratic:

$$c(a) = \gamma a^2.$$

Expected utility is

$$s + (b + \beta\mu)a - \gamma a^2.$$

The worker's best response is

$$a^*(\mu, b, \beta) = \frac{b + \beta\mu}{2\gamma}. \quad (1)$$

The first-best action is

$$a^{FB} = \frac{1}{2\gamma},$$

which is independent of μ .

Interpretation: Objective contracts make a^* move with μ , but the first-best action does not. This is the formal source of dysfunctional behavior.

4 Benchmark models and main theoretical specifications

4.1 Benchmark 1: explicit contract based on the objective measure

With no subjective bonus, the worker's action is

$$a^*(\mu, \beta) = \frac{\beta\mu}{2\gamma}.$$

The optimal explicit piece rate solves:

$$\beta^* = \frac{1}{1 + \text{var}(\mu)}. \quad (2)$$

The firm's expected profit from the optimal explicit contract is

$$V(\beta^*) = \frac{1}{4\gamma[1 + \text{var}(\mu)]} - W_a, \quad (3)$$

where W_a is the worker's alternative wage.

Interpretation: When $\text{var}(\mu)$ is high, the optimal explicit contract uses weak incentives to avoid costly gaming. When $\text{var}(\mu)$ is low, the explicit contract becomes more effective and more profitable.

4.2 Benchmark 2: implicit contract based on subjective performance

With no explicit contract, the firm promises an implicit bonus b when the subjective assessment is favorable. The worker chooses

$$a^*(b) = \frac{b}{2\gamma}.$$

The firm would like to choose $b = 1$, which implements the first-best action. But the firm may renege after observing favorable performance. The self-enforcement constraint is

$$V(b) \geq rb, \quad (4)$$

where r is the firm's discount rate and $V(b)$ is the expected profit stream from maintaining the relationship.

Interpretation: The subjective contract is constrained by reputation. A high bonus creates strong incentives for the worker but also a strong temptation for the firm to refuse payment.

4.3 Combining explicit and implicit contracts: Case 1, profitable explicit fallback

When $V(\beta^*) > 0$, the firm can renege on the implicit contract and fall back to a profitable explicit contract. The reneging constraint is

$$V(b, \beta) - V(\beta^*) \geq rb. \quad (5)$$

For any implicit bonus b , the optimal explicit piece rate is

$$\beta^{**}(b) = \frac{1 - b}{1 + \text{var}(\mu)}. \quad (6)$$

Interpretation: An explicit contract reduces the incentive problem left for the implicit contract. But by improving the firm's fallback position, it also makes the firm more willing to renege. This is why objective contracts can destroy subjective contracts.

4.4 Case 1 comparative statics

The paper shows that implicit incentives are infeasible when the discount rate is sufficiently high or when the objective measure is sufficiently close to perfect. Intuitively:

- higher r makes future reputation less valuable;
- lower $\text{var}(\mu)$ makes the fallback explicit contract more attractive;
- higher W_a reduces the profit available to support relational incentives.

Interpretation: This is the paper's first main result. Better objective measurement can crowd out subjective incentives because it improves the firm's post-reneging outside option.

4.5 Combining explicit and implicit contracts: Case 2, unprofitable explicit fallback

When $V(\beta^*) < 0$, the firm cannot profitably operate using the explicit contract alone. The outside option after reneging is shut-down profit, normalized to zero. The reneging constraint becomes

$$V(b, \beta) \geq rb. \quad (7)$$

In this case, explicit and implicit contracts can be complements. A small explicit piece rate can raise relationship value and thereby make a subjective bonus self-enforcing, while the subjective bonus fixes incentive margins the explicit piece rate cannot handle alone.

Interpretation: This is the paper's second main result. Objective and subjective measures are not always substitutes. The right combination can make production profitable even when each component alone fails.

4.6 Imperfect subjective performance assessment

The benchmark subjective measure y is assumed to be perfect but noncontractible. The paper then relaxes this by allowing a subjective assessment q that is an imperfect proxy for y . The qualitative results remain:

- implicit contracts can still be infeasible when explicit fallback contracts are too good;
- explicit and implicit contracts can still be complements;
- the best mix depends on objective distortion, subjective accuracy, and relational enforcement.

Interpretation: Subjective evaluations need not be perfect to be valuable. They only need to improve incentives relative to the available objective measure, subject to self-enforcement limits.

4.7 Subjective weights on objective measures

In the second model, the firm can subjectively assess μ after performance is observed. If this assessment were contractible, the first-best objective piece rate would be

$$\beta(\mu) = \frac{1}{\mu}. \quad (8)$$

This would undo the distortion in p . Because μ is noncontractible, the firm instead uses an explicit component β plus a subjective component $b(\mu)$.

Interpretation: The firm can use subjective pay not only to reward a separate subjective outcome, but also to adjust how an objective outcome is rewarded. This explains why real-world contracts can specify performance measures clearly but leave pay consequences discretionary.

5 Theoretical identification strategy

5.1 Step 1: identify the objective-measure distortion

The private state μ captures the gap between measured performance and true contribution. If $\text{var}(\mu) = 0$, the objective measure perfectly tracks value. If $\text{var}(\mu)$ is high, the objective measure creates severe gaming incentives.

Interpretation: The model identifies distortion through state-contingent marginal productivity differences, not through statistical noise alone.

5.2 Step 2: identify the self-enforcement constraint

The implicit contract is constrained by the firm's temptation to renege. The temptation depends on the promised bonus and the firm's fallback. The fallback is either the best explicit contract or shut-down.

Interpretation: This is the relational-contract identification step. Subjective performance pay is limited not by courts but by the value of future cooperation.

5.3 Step 3: compare contracts in isolation and in combination

The paper solves three problems:

1. optimal explicit contract only;
2. optimal implicit contract only;
3. optimal combined explicit and implicit contract.

The comparison identifies whether objective and subjective incentives are substitutes or complements.

Interpretation: The result is not monotone. The same objective measure can help the firm use subjective incentives in one parameter region and destroy them in another.

5.4 Step 4: map parameter changes to organizational predictions

The key comparative statics are:

- higher discount rate r weakens implicit contracts;
- higher worker alternative wage W_a weakens implicit contracts;
- lower distortion in the objective measure strengthens explicit contracts but can weaken implicit contracts by improving the fallback;
- greater accuracy of objective measures can increase both explicit and subjective incentive weights when the two are complementary.

Interpretation: These comparative statics connect the model to organizational design: trust, outside options, objective measurement technology, and performance-measure distortion jointly determine compensation design.

6 Figures and detailed interpretations

6.1 Figures I and II: subjective bonus feasibility with discount rate and alternative wage

Read:

- Figure I plots expected profit $V(b)$ against the subjective piece rate b , and overlays reneging lines rb for discount rates $r = 5\%$, 7% , and 10% .
- At low discount rates, the first-best $b = 1$ is feasible. At intermediate discount rates, only a lower subjective bonus is feasible. At high discount rates, no implicit contract is feasible.
- Figure II repeats the logic for alternative wages $W_a = 0.02$ and $W_a = 0.03$.
- A higher alternative wage lowers relationship surplus and therefore lowers the largest self-enforcing subjective bonus.

Interpretation: Figures I and II illustrate the basic repeated-game logic. Subjective pay works only when future relationship value is large enough to make the firm honor the bonus.

Economic message: Trust is not a soft add-on; it is a mathematical constraint. Firms with low future surplus or impatient owners cannot credibly promise large discretionary bonuses.

$\alpha^*(1) = 1/(2\gamma)$ is precisely the first-best action, α^* .

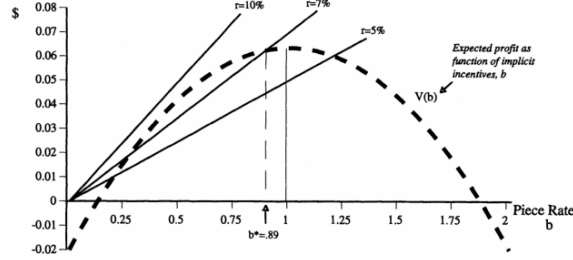


FIGURE I

Figure I shows that the optimal subjective piece-rate declines with the firm's discount rate. The figure depicts the expected profit per period (dashed lines) as a function of implicit incentives $V(b)$ defined by (11) based on $\gamma = 3$ and $w = 0.02$.

(a) Figure I: subjective piece-rate and discount rate.

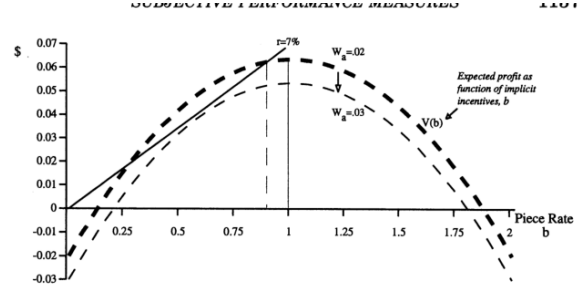


FIGURE II

Figure II shows that the optimal subjective piece-rate declines as the worker's alternative wage increases. The figure, which assumes that $\gamma = 3$, depicts the expected profit per period $V(b)$ at two alternative wages $w = 0.02$ and $w = 0.03$.

(b) Figure II: subjective piece-rate and alternative wage.

6.2 Figures III and IV: substitutability and complementarity between explicit and implicit incentives

Read:

- Figure III shows how the optimal implicit and explicit piece rates vary with the distortion in the objective performance measure, $\text{var}(\mu)$.
- The top panel shows the two benchmark contracts in isolation: the explicit piece rate β^* declines as distortion rises, while the implicit piece rate b^* is unaffected by objective-measure distortion.
- The bottom panel shows the combined contract. When objective measurement is close to perfect, implicit incentives are infeasible. For intermediate and high distortions, implicit incentives become feasible and explicit incentives decline.
- Figure IV shows the complementarity case. With a higher discount rate, neither implicit nor explicit incentives alone can work over a range, but the combined contract can work.

Interpretation: These are the paper's main figures. Figure III shows explicit and implicit contracts as substitutes because a good explicit fallback destroys the firm's credibility. Figure IV shows them as complements because an explicit component can create enough relationship value to sustain an implicit component.

Economic message: A firm should not choose between objective and subjective pay in isolation. The two instruments jointly determine both effort incentives and the credibility of future subjective bonuses.

$w_a = 0.02$, and an effort-disutility parameter $\gamma = 3$. The top panel

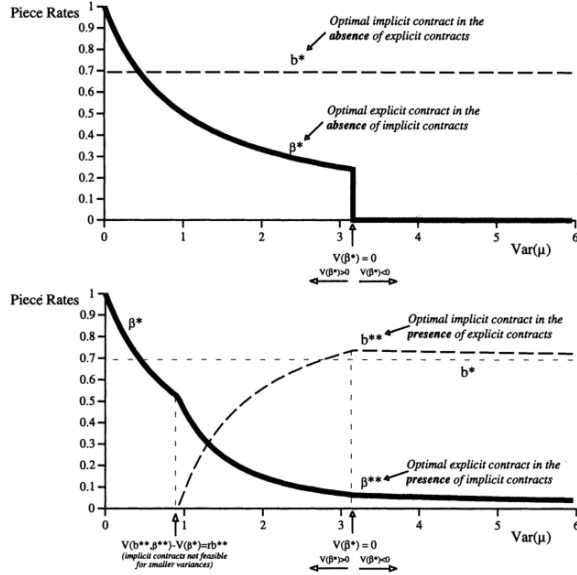


FIGURE III

Figure III shows the interaction between implicit and explicit incentives in optimal incentive contracts. The top panel shows b^* , the bonus for an implicit contract in the absence of any explicit contract, and β^* , the bonus for an explicit contract if no implicit contract is used. The bottom panel shows b^{**} and β^{**} , the implicit and explicit bonuses when the two contracts are used in combination. The

(a) Figure III: interaction of implicit and explicit incentives.

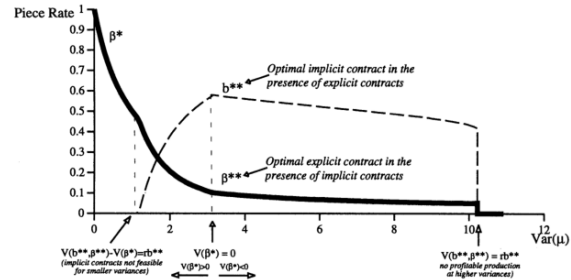


FIGURE IV

Figure IV depicts the case where implicit incentives are not feasible without explicit incentives. The figure assumes that $w_a = 0.02$, $r = 9$ percent, and $\gamma = 3$. No implicit contract is feasible on its own, so b^* (the bonus for an implicit contract in the absence of any explicit contract) is not defined. If $\text{var}(\mu) > 10.3$, implicit contracts are infeasible even in combination with the optimal explicit contract, so

(b) Figure IV: complementarity of implicit and explicit incentives.

6.3 Figures V and VI: subjective weights on objective performance measures

Read:

- Figure V shows the first-best combination of explicit and implicit weights on an objective performance measure.
- If μ were contractible, the firm would set $\beta(\mu) = 1/\mu$ to neutralize measurement distortions.
- Because μ is subjective, only part of this state-contingent reward can be supported by reputation.
- Figure VI shows the second-best contract. The firm uses more explicit incentives than the first-best in order to reduce the temptation to renege on high subjective payments.

Interpretation: These figures explain why firms may deliberately use vague bonus systems. The objective measure is known, but the weight placed on it is adjusted subjectively to reflect conditions that cannot be written into court-enforced contracts.

Economic message: Subjective pay can correct objective-measure gaming. But because subjective rewards are limited by self-enforcement, the firm may still use too much objective incentive relative to the first-best.

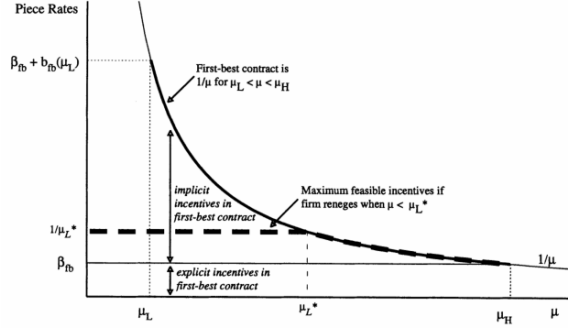


FIGURE V

Figure V shows the first-best combination of explicit and implicit weights on the objective performance measure (bold curve), as well as the maximum feasible total incentives associated with first-best explicit incentives (bold dashed line). See text for definition of variables.

(a) Figure V: first-best subjective weights.

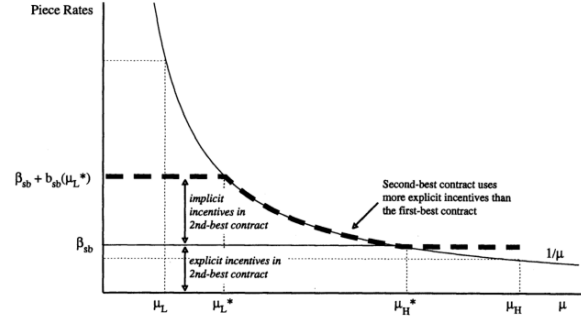


FIGURE VI

Figure VI shows the second-best combination of explicit and implicit weights on the objective performance measure (bold dashed line). See the text for the definition of variables.

(b) Figure VI: second-best subjective weights.

7 Interpretive synthesis

7.1 Why objective measures can be too good

A better objective measure seems unambiguously desirable in static agency theory. This paper shows why that is not always true in a relational setting. If explicit contracts are too effective, the firm has a valuable fallback after reneging on a subjective bonus. The value of the ongoing relationship falls, and subjective contracting may become infeasible.

7.2 Why hybrid incentive systems are common

Hybrid systems combine enforceability and discretion. Objective pay provides a baseline incentive and reduces the size of discretionary rewards. Subjective pay adjusts for missing dimensions of true contribution and mitigates gaming. The optimal mix depends on measurement distortion, reputation value, and the worker's outside option.

7.3 Why subjective pay is not simply favoritism

The model treats subjective assessment as economically useful information. It can capture quality, cooperation, innovation, long-run customer effects, and state-contingent difficulty. Favoritism and supervisor bias are possible extensions, but the core model shows that subjectivity can be productive even without bias.

7.4 Connection to firm boundaries

The model gives a relational-contract version of the Coasean theory of the firm. Firms are valuable where spot-market explicit contracts are imperfect. But if spot contracts become too good, the firm may lose its distinctive ability to sustain implicit contracts.

8 Short summary

- Objective performance measures are enforceable but distortionary.

- Subjective performance assessments can better capture true contribution but require self-enforcement.
- The optimal explicit piece rate is $\beta^* = 1/[1 + \text{var}(\mu)]$.
- The subjective bonus is feasible only if relationship value exceeds the temptation to renege.
- Good explicit contracts can destroy implicit contracts by improving the firm's reneging fallback.
- Explicit and implicit contracts can also be complements when neither one alone generates enough profit.
- Subjective weights on objective measures allow firms to correct dysfunctional behavior induced by objective piece rates.
- The model explains why real compensation systems combine objective pay, subjective bonuses, and deliberately vague reward formulas.

9 Conclusion

9.1 Core conclusion

Baker, Gibbons, and Murphy show that subjective performance measures are not a residual category outside economic analysis. They are central to optimal incentive design when objective measures distort effort and subjective assessments require trust.

9.2 Economic intuition

Objective contracts solve the enforcement problem but create a measurement problem. Subjective contracts solve part of the measurement problem but create an enforcement problem. Optimal compensation balances these two frictions, sometimes using explicit and implicit incentives as substitutes and sometimes as complements.

9.3 Incremental contribution in one sentence

The paper's distinctive contribution is to show that the interaction between explicit and implicit incentives is itself a design problem: objective contracts can either crowd out or support subjective contracts depending on how they affect effort distortions and the firm's temptation to renege.

9.4 Final takeaway

The paper provides a formal foundation for a central fact of organizations: firms often pay for what they can measure, but they must also reward what they can only judge. The challenge is not whether to use objective or subjective performance measures, but how to combine them so that effort is directed toward long-run firm value.